Probability and Random Processes
Assignment 2

Date: May 02, 2026
Course Instructor:
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Spring 2025

Total Marks: 100
Total Questions: 18
Submission: Group Assignment

Group No.	Student 1	Student 2	Student 3

Important Note: This assignment must be completed in groups of three students. A viva will be conducted for this assignment, and full marks will be awarded only after successful completion of the viva.
If any student fails to perform satisfactorily in the viva, that student will receive zero marks for the assignment. The viva will be conducted in the respective groups. The time slot for each group will be shared.

Problem 1 [5 marks]

The performance scores of employees at a software company follow a normal distribution with mean 78 and standard deviation 12, based on evaluations from 800 employees.

(a) What is the probability that an employee's performance score is between 75 and 90? Approximately how many employees received a score in this range?

(b) What is the probability that a randomly selected employee has a performance score greater than 85?

(c) What is the probability that an employee's performance score is less than 70?

(d) Determine the performance score that corresponds to the top 5% of employees.

(e) The middle 30% of performance scores fall between what two values?

Problem 2 [6 marks]

A stick of length 10 units is broken at a random point X, uniformly distributed between 0 and 10. The piece of length X is then broken at a point Y, uniformly distributed between 0 and X. Define

$$Z = X - Y.$$

Find the conditional expectation

$$E[X \mid Z].$$

Problem 3 [4 marks]

Sohail Abbas works at Apple's customer support center, where he handles two types of queries: iPhone-related issues and MacBook-related issues. Based on historical data, the support center receives an average of 6 iPhone-related queries per hour and 4 MacBook-related queries per hour. The number of queries for each category follows an independent Poisson distribution.

(a) What is the probability that Sohail receives exactly 12 queries in an hour?

(b) Given that Sohail received a total of 8 queries in an hour, what is the probability that 5 of them were iPhone-related?

(c) If Sohail works an 8-hour shift, what is the expected total number of queries he handles in a day?

Hint: The sum of two independent Poisson random variables is also a Poisson random variable.

Problem 4 [4 marks]

How many times would you expect to roll a fair die before all 6 sides appeared at least once?

Problem 5 [4 marks]

The time X, in hours, required to repair a machine is exponentially distributed with parameter

$$\lambda = \frac{1}{4}.$$

(a) Compute the probability that the repair time exceeds 3 hours.

(b) Find the conditional probability that a repair takes at least 12 hours, given that the repair time has already exceeded 4 hours.

(c) Given that the repair time has already exceeded 4 hours, find the expected additional time required to complete the repair.

(d) Find the probability that the total repair time falls between 4 and 16 hours, that is,

$$P(4 < X \leq 16).$$

Problem 6 [6 marks]

Mohammad Rizwan is playing a series of cricket matches. Each match is either a win with probability p or a learn/lose with probability $1 - p$. He plans to play 5 matches, but if he wins the fifth match, then he will continue playing until he loses and learns.

(a) Find the expected number of matches that Rizwan plays.

(b) Find the expected number of matches that he loses.

(c) Let

$$X \sim \text{Negative Binomial}(r, p),$$

where X represents the number of failures before the r-th success in a sequence of independent Bernoulli trials with success probability p. Find the value of p that maximizes $P(X = k)$ for a fixed k.

Problem 7 [5 marks]

Let X be a normal random variable with parameters

$$\mu = 5, \qquad \sigma^2 = 17,$$

and let

$$Y = \frac{X - 5}{\sqrt{17}}.$$

Compute the following probabilities:

(a) $P(X > 4)$

(b) $P(3 < X < 7)$

(c) $P(X < 2)$

(d)

$$P\left(X > 5 + \frac{Y^2}{2}\right)$$

Problem 8 [5 marks]

Aleena is buying a car. She finds information online which states that the total number of thousands of km that a car can be driven before it is discarded is an exponential random variable with

$$\lambda = \frac{1}{15}.$$

Ali is selling his old car, which he says has been driven only 10,000 km.

(a) If Aleena buys his car, what is the probability that she would get at least 20,000 additional km out of it?

(b) Now assume that the lifetime mileage of the car, in thousands of km, is uniformly distributed over $(0, 50)$ instead. Find the probability that she would get at least 20,000 additional km out of it, given that the car has already been driven 10,000 km.

Problem 9 [4 marks]

A die is rolled 12,000 times and a 6 is obtained 3,000 times. Is this a fair die? Clearly show your working.

Problem 10 [6 marks]

Suppose an employee working in a nuclear plant has a hazard function of getting cancer given by

$$\lambda(y) = 0.065 + \frac{0.034 \ln(y)}{y^2},$$

where y is the number of years he has been exposed to radiation.

(a) What is the probability that a worker who has already been exposed to 30 years of radiation lives to the age of 45?

(b) How does the probability from part (a) compare to the probability that he survives to age 51? What is the change in probability?

Problem 11 [5 marks]

On a given day, the total number of people entering a post office is a Poisson random variable,

$$N \sim \text{Poisson}(\lambda).$$

Each arriving person is male with probability p and female with probability $1-p$, independently of all other individuals.

Let M be the number of males and F be the number of females entering the post office on that day.

(a) Find the distribution of M.

(b) Find the distribution of F.

(c) Show whether M and F are independent.

Problem 12 [5 marks]

Let X be a continuous random variable uniformly distributed on the interval $[0, 2]$, and let Y be an exponential random variable with rate parameter $\lambda = 1$, independent of X. Define a new random variable

$$Z = XY.$$

(a) Derive the conditional joint density function of X and Y given $X > 1$.

(b) Using your result from part (a), evaluate the conditional expectation

$$E[Z \mid X > 1].$$

Problem 13 [6 marks]

A Gaussian random variable has a pdf of the form

$$f_X(x) = \sqrt{\frac{2}{\pi}} e^{-2(x+1)^2}.$$

Compute the following probabilities in terms of the Q-function with positive arguments and also give numerical evaluations:

(a)
$$P\left(|X+1| > 3 \ \cup \ |X-1| < 2\right)$$

(b)
$$P\left(|X+1| > 3 \mid |X-1| < 2\right)$$

Problem 14 [8 marks]

Consider a bulb factory that manufactures two types of bulbs: Long-life and Short-life. We know that the factory manufactures 3 times more Long-life bulbs compared to Short-life bulbs. The conditional lifetimes for the two types of bulbs are given by

$$f_{X|S}(x) = \frac{1}{100} e^{-x/100} u(x),$$

and

$$f_{X|L}(x) = \frac{1}{1000} e^{-x/1000} u(x).$$

Suppose we are given a bulb without knowing its type. To determine, we turn on the bulb and let it operate until it fails, giving us an observation of the random variable X. We devise a Bayesian method to determine the bulb type from this observation.

(a) Determine a threshold γ, such that when $X < \gamma$, your Bayesian decision is a Long-life bulb, and when $X > \gamma$, your Bayesian decision is a Short-life bulb.

(b) For the decision rule in part (a), determine the probability that your decision will be wrong.

(c) Suppose you observe that the bulb you are testing is still working after 10 hours of operation. Given this observation, determine the probability that it will fail in the next 20 hours.

Problem 15 [6 marks]

Consider a robot, the position of which is uniformly distributed in a circle of radius 2 meters. Let X and Y represent the random coordinates.

(a) Determine the probability that the robot lies inside a circle of radius r, for any $0 \leq r \leq 2$.

(b) Let

$$\Theta = \tan^{-1}\left(\frac{Y}{X}\right)$$

be measured in radians. Determine the probability of the event

$$\{\Theta \leq \theta\}$$

for any $0 \leq \theta \leq 2\pi$.

Hint: Change coordinates to polar coordinates and use the Jacobian.

(c) Determine the average distance of the robot from the origin.

(d) Determine

$$E[\Theta].$$

Problem 16 [6 marks]

Let X correspond to the random waiting time at the Less-than-10-items counter of a supermarket. Suppose we know that X is an exponential random variable with mean 2 minutes.

Let Y be the random waiting time at the regular counter, which is also known to be exponentially distributed but with mean 5 minutes. If X and Y are known to be statistically dependent, determine the probability that a customer will have to wait for a longer time at the Less-than-10-items counter than she will have to wait at the regular counter.

Consider a pair of random variables X and Y that are uniformly distributed within a square region, that is,

$$f_{X,Y}(x,y) = \begin{cases} C, & |x| \leq 1,\ |y| \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Determine the constant C.

(b) Show that X and Y are independent random variables.

(c) Determine the probability that

$$X^2 + Y^2 \leq r^2$$

for all $r \geq 0$.

Problem 17 [8 marks]

Consider the optical communication system consisting of two optical transmission paths. For the transmission of bit-0, both lasers are turned off for a duration of T seconds, while for the transmission of bit-1, both lasers are turned on for the same duration.

Let X_1 be the random variable representing the number of electrons counted at the output of Photodetector 1 during a T-second interval. Similarly, let X_2 represent the number of electrons counted at the output of Photodetector 2.

Both X_1 and X_2 have identical distributions. They are Poisson distributed with mean R_0 when the corresponding laser is turned off, and Poisson distributed with mean R_1, where $R_1 > R_0$, when the corresponding laser is turned on.

Assume that the bits are equally likely. Also assume that, given the knowledge of the bit, X_1 and X_2 are statistically independent.

(a) Suppose you have the observation

$$X_1 = k_1, \qquad X_2 = k_2,$$

and you need to decide whether bit-1 or bit-0 was transmitted. What condition should k_1 and k_2 satisfy for you to decide that bit-0 was transmitted?

(b) Evaluate the probability of error associated with your decision rule in part (a). Simplify as much as possible.

Problem 18 [7 marks]

An environmental engineer is studying how the outside air temperature affects the hourly cooling energy used by a small data center. During several test periods, all other operating conditions were kept approximately constant. The following data were recorded.

Cooling energy y (kWh)	Temperature x (°F)	Cooling energy y (kWh)	Temperature x (°F)
180	30	201	37
194	48	224	63
231	70	207	41
218	57	236	76

(a) Draw a scatter plot of the data, with temperature x on the horizontal axis and cooling energy y on the vertical axis.

(b) Estimate the least-squares regression line

$$\hat{y} = a + bx.$$

Find the slope b and intercept a.

(c) Use the fitted regression line to predict the cooling energy when the outside temperature is

$$x = 65°\text{F}.$$